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PARAMETER IDENTIFICATION OF VISCOELASTIC MATERIALS USING DIFFERENT DEFORMATION VELOCITIES

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Abstract:

In this paper the fractional Kelvin-Zener model of viscoelastic body was used for modeling the behavior of monofilament and fluorocarbon during the ramp and hold type of experiment, using different deformation velocities. First, experimental measurements were made on 9 samples of each material. Three sample groups were tested at different deformation rates during the first phase of the experiment. Afterward, the relaxation phase was kept consistent across all samples. Riemann-Liouville fractional derivatives were approximated numerically using the Grünwald-Letnikov definition. Four model parameters have been determined for each sample by numerical procedure. Finally, the experimental results were compared with the results provided by the fractional Kelvin-Zener model.

Key words: viscoelasticity, stress relaxation experiment, fractional derivative.

1. Introduction

Over the past few decades, polymers have gained popularity as a preferred engineering structural material due to their affordability, ease of processing, weight savings, corrosion resistance, and other notable benefits. The use of polymeric materials is ubiquitous, both in our daily lives and in the engineering profession. Viscoelastic properties of polymer fibers has attracted considerable attention in different scientific areas. Experimental studies conducted with different polymers revealed rheological behavior which manifests non-linear elasticity, and almost ideal plasticity, or plasticity with strain hardening, strain-rate effects, creep and viscoelastic relaxation effects. Those kinds of properties of polymer materials play an important role in their application. One of the most important representatives of polymeric materials nylon 6,6 is characterized with high mechanical tensile strength, elasticity, and luster while being wrinkle proof and highly resistant to abrasion [1].

Polymeric materials prolene, ethylene (nylon) and Ticron, otherwise threads that are usually used in plastic surgery, were subjected to tests of time-dependent mechanical properties

responsible for the quality of tendon repair flexors in terms of crack formation also by experiments crawling. Due to its specific viscoelastic properties, Ticron proved to be the most suitable [2].

Monofilament made of fishing nylon has attracted special attention in medicine due to its mechanical properties. A case-control study of sensitivity, specificity, and comparison with standardized sensory modalities of a homemade monofilament for screening of diabetic peripheral neuropathy. A low cost instrument has been proven to be equivalent to the internationally accepted instrument in assessing the risk of foot ulcers in diabetes patients, and that it has the potential to become standard for detecting diabetic neuropathy [3]. As an advantage of fishing nylon in the detection of diabetic neuropathy in a case-control study in addition to its mechanical features, a feature that stood out was its economy, as well as its simplicity and the speed of conducting the test, which shows its potential in daily clinical practice [4].

It was demonstrated that inexpensive high-strength polymer fibers used for fishing line and sewing thread can be easily transformed by twist insertion to provide fast, scalable, nonhysteretic, long-life tensile and torsional muscles. Extreme twisting produces coiled muscles that can contract by 49%, lift loads over 100 times heavier than can human muscle of the same length and weight, and generate amount of mechanical work per kilogram of muscle weight, similar to that produced by a jet engine [5].

The identification of model parameters for viscoelastic materials has been conducted in numerous studies, see for example [6], [7] and [8]. The values of model parameters determined on the bases of relatively simple experiments of stress relaxation and creep are sometimes used for modeling in collision problems, where deformation rates are much higher. This paper investigates the influence of deformation rate on the values of parameters of the fractional Kelvin-Zener model.

2. Experiment and parameter identification

2.1 Stress-relaxation experiment

Eighteen different samples of polymer materials, i.e. nine fluorocarbon (FC) samples with a thickness 0.3 mm, linear density $1.116 \cdot 10^{-4}$ kg/m and nine monofilament (MF) samples with a thickness 0.28 mm, linear density $7.078 \cdot 10^{-5}$ kg/m, were tested with the test system Instron 2710-204.



Fig. 1. Experimental apparatus.

All samples had an initial length of 30 cm. The experimental apparatus is shown in Figure 1. During the first phase, the ramp phase, the samples were loaded under constant strain rate until a predetermined strain value of 10% was achieved. Three sample groups were subjected to different deformation rates, specifically 60 mm/min, 120 mm/min, and 360 mm/min, so that the tension phase lasted 30 s, 15 s, and 5 s, respectively, for these three sample groups. After that, the samples were exposed to a relaxation phase in which the achieved deformation was held constant for a period of 180 s for all samples. Figure 2 shows the experimental results for 9 samples of fluorocarbons, which were grouped into sets of three samples each. The experimental results for the monofilament samples are shown in Figure 3.

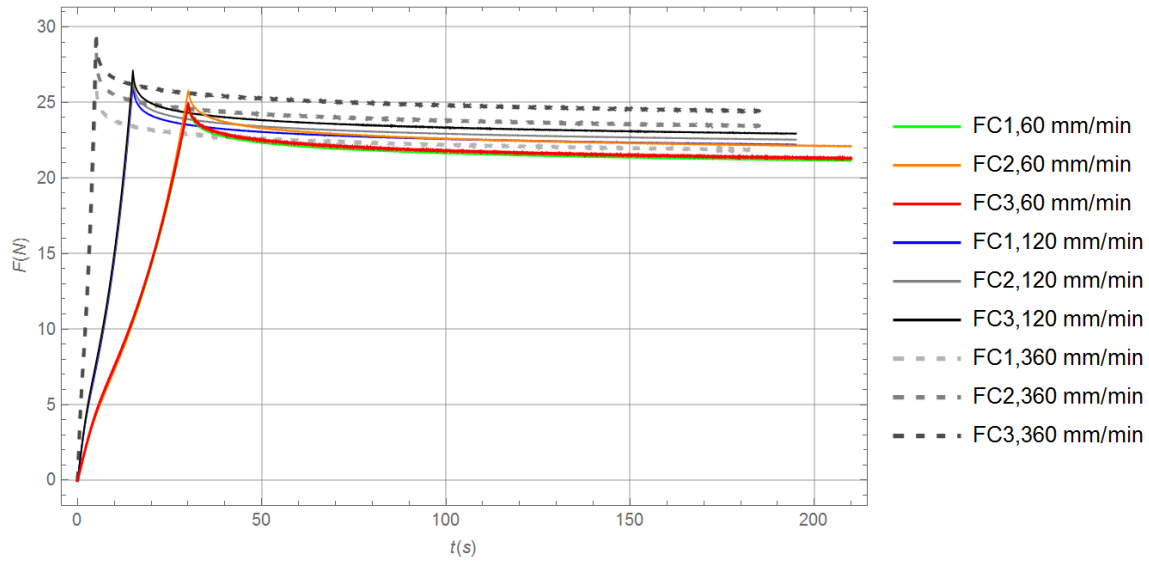


Fig. 2. Experimental results for nine fluorocarbon samples that were subjected to different deformation rates.

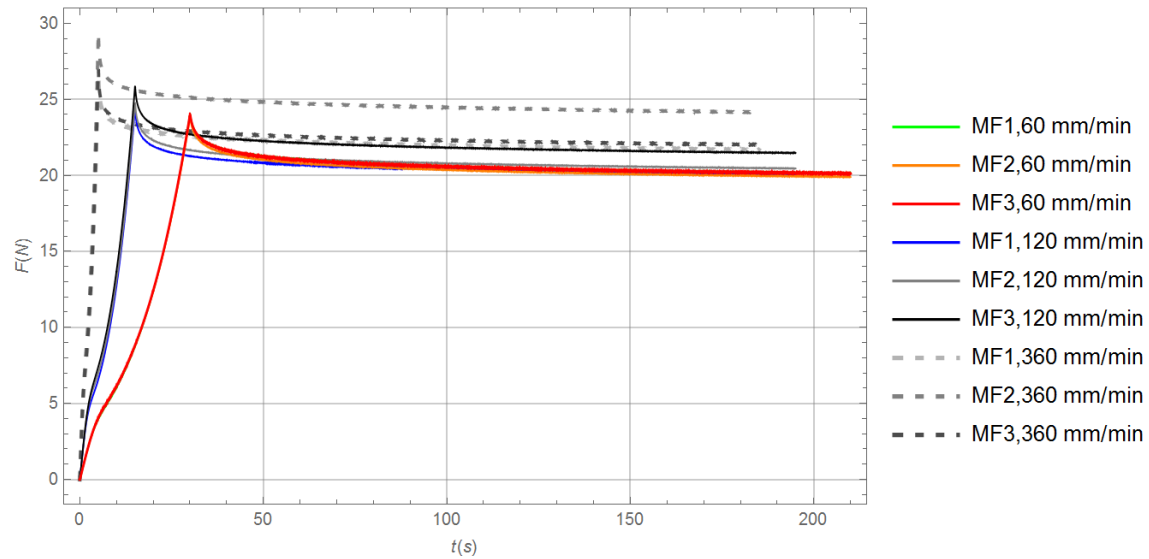


Fig. 3. Experimental results for nine monofilament samples that were subjected to different deformation rates.

By observing the obtained experimental results, certain mutual deviations of the curves within the same group of three samples can be noticed, but from a phenomenological point of view, there is no difference in the behavior of the material during the experiment. From Figure 3, which shows the behavior of the monofilament samples, very good agreement of results for all three samples can be observed at a deformation rate of 60 mm/min. At higher deformation rates, two curves in each group are close, while one curve in each of these groups deviates slightly. The reason for the observed deviation could be the imperfections of the tested samples.

2.2 The model

The appropriate selection of rheological model is important for testing viscoelastic materials. Considering that stress is directly related to the zeroth derivative of strain in solids and to the first derivative of strain in fluids, it is reasonable to hypothesize that in viscoelastic materials, which exhibit characteristics of both solids and fluids, stress could be proportional to the α th derivative of strain, where α is a real number between 0 and 1, see [10]. In this analysis we chose the Fractional Kelvin-Zener model which yields accurate results that match experimental data while using only four parameters. The constitutive equation has the following form:

$$f + \tau_{f\alpha} f^{(\alpha)} = E(x + \tau_{x\alpha} x^{(\alpha)}), \quad (1)$$

where f and x stand for tensile force and elongation, $\tau_{f\alpha}$, $\tau_{x\alpha}$ are model parameters expressed in s^α , E is expressed in N/m, while the real number α ($0 < \alpha < 1$) represents the derivative order in the standard Riemann-Liouville form. Parameters $\tau_{f\alpha}$, $\tau_{x\alpha}$ and E must satisfy certain restrictions

$$E > 0, \tau_{f\alpha} > 0, \tau_{x\alpha} > \tau_{f\alpha}, \quad (2)$$

according to the second law of thermodynamics, [9]. Our objective is to calculate the values of the model parameters (α , E , $\tau_{x\alpha}$, $\tau_{f\alpha}$) for all samples and examine how changes in deformation velocity affect the obtained results.

3. Results

Following the procedure for parameter identification described in [7], where the Grünwald-Letnikov definition of fractional derivatives was used, the values of parameters of the viscoelastic model are determined for all samples, see also [10]. The obtained results are shown in Table 1 for fluorocarbon samples, and in Table 2 for monofilament samples.

Sample	α	$E[\text{N/mm}]$	$\tau_{x\alpha}[\text{s}^\alpha]$	$\tau_{f\alpha}[\text{s}^\alpha]$
FC1 (60 mm/min)	0.527	0.681	1.127	$4.882 \cdot 10^{-4}$
FC2 (60 mm/min)	0.529	0.712	1.104	$4.957 \cdot 10^{-4}$
FC3 (60 mm/min)	0.515	0.684	1.088	$4.958 \cdot 10^{-4}$
FC1 (120 mm/min)	0.437	0.71	0.671	$4.783 \cdot 10^{-4}$
FC2 (120 mm/min)	0.436	0.721	0.679	0.011
FC3 (120 mm/min)	0.447	0.734	0.862	0.153
FC1 (360 mm/min)	0.434	0.712	0.437	$2.971 \cdot 10^{-3}$
FC2 (360 mm/min)	0.43	0.763	0.431	$4.912 \cdot 10^{-4}$
FC3 (360 mm/min)	0.426	0.795	0.433	$4.912 \cdot 10^{-4}$

Table 1. Model parameter values for fluorocarbon at deformation rates of 60, 120, and 360 mm/min.

Sample	α	$E[\text{N/mm}]$	$\tau_{x\alpha}[\text{s}^\alpha]$	$\tau_{f\alpha}[\text{s}^\alpha]$
MF1 (60 mm/min)	0.585	0.652	1.463	$4.888 \cdot 10^{-4}$
MF2 (60 mm/min)	0.583	0.643	1.479	10^{-3}
MF3 (60 mm/min)	0.598	0.652	1.533	$4.962 \cdot 10^{-4}$
MF1 (120 mm/min)	0.513	0.648	0.898	$4.949 \cdot 10^{-4}$
MF2 (120 mm/min)	0.517	0.661	0.893	$4.95 \cdot 10^{-4}$
MF3 (120 mm/min)	0.514	0.694	0.86	$1.798 \cdot 10^{-5}$
MF1 (360 mm/min)	0.525	0.714	0.505	$2.466 \cdot 10^{-4}$
MF2 (360 mm/min)	0.532	0.797	0.481	$4.974 \cdot 10^{-4}$
MF3 (360 mm/min)	0.533	0.725	0.507	$4.971 \cdot 10^{-4}$

Table 2. Model parameter values for monofilament at deformation rates of 60, 120, and 360 mm/min.

From the obtained results, it can be noticed that there are no significant deviations in the parameter values within each group of three samples for both tested materials, indicating result repeatability. By observing the obtained results for the derivative order α , it can be concluded that the values of α at a deformation rate of 60 mm/min are slightly higher for both materials compared to the values obtained at higher speeds. With the increase of the deformation rate, there is a slight increase in the values of the parameter E . The biggest change in the parameter values is observed in the parameter $\tau_{x\alpha}$, where the values decrease with the increase of the deformation rate. The values of the parameter $\tau_{f\alpha}$ are very small for most of the samples compared to the obtained values of other parameters. Considering that this parameter multiplies the α th derivative of the tensile force in equation (1), it can be concluded that the influence of this term is reduced.

4. Conclusions

In this paper we analyzed the influence of strain rate on the value of parameters of the fractional Kelvin-Zener viscoelastic model in a stress relaxation experiment. The behavior of two viscoelastic materials was tested, with three sample groups formed for each material, all subjected to the same deformation speed. Very similar behavior was observed within each sample group, as evidenced by the small deviations in the parameter values obtained for each group." The change in deformation speed affected the parameter values in a very similar manner for both materials. The highest and lowest deformation speeds for the tested samples were in a 6:1 ratio. It would be interesting to investigate the effect of a much larger ratio between the maximum and minimum deformation speeds, and to analyze the potential application of relatively simple relaxation and creep experiments to collision problems where deformation speeds are very high.

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